Calculating probabilities

This example shows how to work out the probability of randomly picking a red candy from a group of 10 candies. The number of ways this event could happen is put at the top of the fraction and the total number of possible events is put at the bottom.

number of red candies that can be chosen

$$\frac{3 \text{ red}}{10 \text{ candies}}$$

total that can be chosen

chance of red candy being chosen, as fraction

$$\frac{3}{10} \text{ or } 0.3$$

chance of red candy being chosen, as decimal

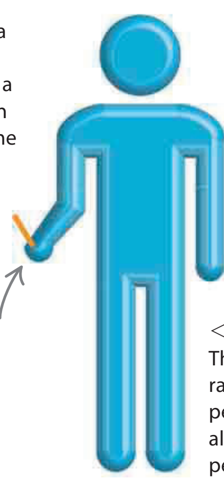
△ **Pick a candy**
There are 10 candies to choose from. Of these, 3 are colored red. If one of the candies is picked, what is the chance of it being red?

△ **Red randomly chosen**
One candy is chosen at random from the 10 colored candies. The candy chosen is one of the 3 red candies available.

△ **Write as a fraction**
There are three reds that can be chosen, so 3 is put at the top of the fraction. As there are ten candies in total, 10 is at the bottom.

△ **What is the chance?**
The probability of a red candy being picked is 3 out of 10, written as the fraction 3⁄10, the decimal 0.3, or the percentage 30%.

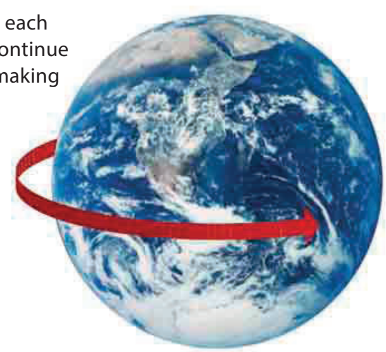
◁ **Heads or tails**
If a coin is tossed there is a 1 in 2, or even, chance of throwing either a head or a tail. This is shown as 0.5 on the scale, which is the same as half, or 50%.



majority of people are right-handed

◁ **Being right-handed**
The chances of picking at random a right-handed person are very high—almost 1 on the scale. Most people are right-handed.

▷ **Earth turning**
It is a certainty that each day the Earth will continue to turn on its axis, making it a 1 on the scale.



0.5
EVEN CHANCE

LIKELY

1
CERTAIN